

TECHNICAL OCCUPATIONAL



WELDING

Y4/T1

Learner Name:..... Surname:.....

SAFETY PROCEDURES

1. Emergency Response Plan



- Know emergency contact numbers
- Identify nearest hospital or medical facility
- Have a first aid kit nearby

2. Hazard Identification

- Recognize potential hazards (e.g., wildlife, terrain, weather)
- Assess risks and take precautions

RISK ASSESSMENT

1. Identify potential hazards: Think about the activities you'll be doing, the environment you'll be in, and the equipment you'll be using.
2. Assess the likelihood and impact: Consider how likely each hazard is to occur and how severe the consequences could be.
3. Prioritize risks: Focus on the most critical risks that require immediate attention.

PRECAUTIONS

1. Research and planning: Gather information about the location, weather, and potential hazards.
 2. Equipment and gear: Ensure you have the necessary equipment and gear to mitigate risks.
 3. Safety protocols: Establish clear safety protocols, such as emergency contact information and a communication plan.
3. Personal Safety
- Stay alert and aware of surroundings
 - Use buddy system when exploring

BENEFITS OF THE BUDDY SYSTEM

1. Safety in numbers: When you're with a buddy, you can look out for each other in case of an emergency.
2. Shared knowledge: Two heads are better than one. Your buddy may have knowledge or skills that can help you navigate challenging situations.

3. Moral support: Exploring can be physically and mentally demanding. Having a buddy can provide emotional support and motivation.
4. Emergency response: If one of you gets injured or lost, the other can provide assistance or go for help.

BEST PRACTICES FOR THE BUDDY SYSTEM



1. Choose a compatible buddy: Select someone with similar experience, skills, and physical abilities.
2. Communicate before heading out: Discuss your plans, including the route, expected duration, and emergency procedures.
3. Stay together: Make sure to stay within sight or earshot of each other, especially in challenging terrain.
4. Establish a communication plan: Decide on a communication method, such as two-way radios or a whistle signal, in case you get separated.
5. Review and adjust: After each excursion, discuss what worked well and what didn't, and make adjustments for future trips.

II. FIRST AID BOXES

1. Basic First Aid Kit Contents:



Wound Care

1. Bandages (assorted sizes): For covering and protecting wounds.
2. Gauze pads (assorted sizes): For applying pressure, cleaning wounds, and absorbing bleeding.
3. Gauze roll: For wrapping injuries, such as sprains or strains.
4. Medical tape: For securing bandages or gauze in place.

Wound Cleaning and Disinfecting

Disinfecting Cuts and Scrapes



1. Antiseptic wipes: For cleaning and disinfecting wounds.
2. Antibiotic ointment: For applying to wounds to prevent infection.

Pain Relief and Antihistamines

1. Acetaminophen (such as Tylenol): For relieving pain and reducing fever.
2. Ibuprofen (such as Advil or Motrin): For relieving pain, reducing fever, and reducing inflammation.
3. Benadryl (diphenhydramine): For relieving allergic reactions, itching, and hives.

Blister and Burn Care



1. Aloe vera gel: For soothing and treating minor burns.
2. Burn cream: For treating minor burns and scalds.
3. Moleskin: For protecting and cushioning blisters.

General Medical

1. First aid manual: For providing guidance on basic first aid techniques.
2. Medical gloves: For protecting against bodily fluids and preventing infection.
3. Scissors: For cutting bandages or clothing.
4. Tweezers: For removing splinters or ticks.
5. Thermometer (oral or digital): For checking body temperature.
6. Eye wash solution: For flushing out debris or chemicals from the eyes.
7. Triangular bandage: For immobilizing injured limbs or creating a sling.
8. Cold pack: For reducing swelling and relieving pain.
9. Snake bite kit (if necessary): For treating snake bites in areas where venomous snakes are common.
10. Over-the-counter medications (such as anti-diarrheal or motion sickness medication): For treating specific ailments.

2. First Aid Techniques:
 - *Wound cleaning and dressing*

step-by-step guide:

Wound Cleaning

1. *Wash your hands: Before touching the wound, wash your hands thoroughly with soap and water.*
2. *Rinse with saline solution: Use a saline solution (0.9% sodium chloride) to rinse the wound and remove any debris.*
3. *Clean with mild soap: Gently clean the wound with a mild soap, taking care not to scrub or use harsh soap.*
4. *Rinse thoroughly: Rinse the wound with saline solution or sterile water to remove any soap residue.*

Wound Dressing

<https://youtu.be/woRcSlidK0I>

1. *Pat dry:*
Gently pat the wound dry with a clean towel.
2. *Apply antibiotic ointment:*
Apply a thin layer of antibiotic ointment to the wound to help prevent infection.
2. *Cover with dressing:*
Cover the wound with a sterile dressing, such as gauze or a bandage.
4. *Secure dressing: Secure the dressing with medical tape, taking care not to tape too tightly.*
5. *Change dressings regularly:*
Change the dressing daily or as needed to promote healing and prevent infection.
6. *Monitor for infection:*
a) Monitor the wound for signs of infection, such as redness, swelling, increased pain, or pus.
7. *Seek medical attention:*
a) Seek medical attention if you notice any signs of infection or if the wound is not healing as expected.

BLEEDING CONTROL

Bleeding control is a critical first aid skill that can help save lives. Here's a step-by-step guide on how to control bleeding:

External Bleeding

1. *Put on gloves:*
If available, put on medical gloves to protect yourself from bodily fluids.
2. *Apply pressure:*
a) Use a clean cloth or gauze to apply firm and steady pressure to the wound.
Elevate the injured limb above heart level if possible.
3. *Use a tourniquet (if trained):*
a) If the bleeding is severe and you are trained to use a tourniquet, apply it according to the manufacturer's instructions.

4. Apply a hemostatic agent (if available):
 - a) Hemostatic agents, such as QuikClot, can help promote clotting and control bleeding.

Internal Bleeding

1. Call for medical help:

Internal bleeding is a life-threatening condition that requires immediate medical attention.

2. Keep the person calm:

Try to keep the person calm and comfortable until medical help arrives.

3. Monitor vital signs:

Keep an eye on the person's vital signs, such as pulse, breathing rate, and blood pressure.

4. Do not attempt to treat internally:

Do not attempt to treat internal bleeding yourself, as this can cause more harm than good.

5. Stay calm:

It's essential to remain calm when dealing with bleeding, as this will help you think more clearly and make better decisions.

6. Use clean equipment:

Always use clean equipment and supplies when treating bleeding to minimize the risk of infection.

7. Apply pressure for at least 5 minutes:

When applying pressure to a wound, make sure to do so for at least 5 minutes to allow for clotting to occur.

8. Do not remove embedded objects:

If an object is embedded in the wound, do not attempt to remove it. Instead, apply pressure around the object and seek medical attention immediately.

CARDIOPULMONARY RESUSCITATION (CPR)



Cardiopulmonary resuscitation (CPR) is a lifesaving technique that combines chest compressions with rescue breaths to help maintain blood circulation and oxygenation to the brain and other vital organs. Here's a step-by-step guide:

Adult CPR

1. Call 10111 or your local emergency number: If you are alone with the person, call 911 or your local emergency number before starting CPR. If you are with someone else, have one person call while the other starts CPR.
2. Check the person's airway, breathing, and circulation (ABCs): Position the person on their back on a firm, flat surface. Tilt their head back and lift their chin to open their airway. Listen for breath sounds and feel for air on your cheek. Check for a pulse in the neck.
3. Start chest compressions: If the person is unresponsive and not breathing, begin chest compressions. Place the heel of one hand on the center of the person's chest, just above the nipple line. Place your other hand on top of the first hand, with your fingers interlaced. Push down on the chest with the heel of your hand to a depth of 2-3 inches(5-7cm). Release the pressure, allowing the chest to return to its normal position. Repeat at a rate of 100-120 compressions per minute.
4. Give rescue breaths: After 30 chest compressions, give two breaths. Pinch the person's nose shut and give one breath through their mouth. The breath should last one second and make the person's chest rise. Give a second breath. Return to chest compressions.
5. Continue CPR: Continue CPR until emergency medical personnel arrive, the person starts showing signs of regaining consciousness, or you are physically unable to continue.

III. Protective Clothes and Colour codes

Protective clothes, also known as personal protective equipment (PPE), are essential for ensuring safety in various workplaces and environments. Here's a list of common protective clothes and their corresponding color codes:

HEAD PROTECTION



1. Hard Hats: Protect the head from falling objects, electrical hazards, and other impacts.
 - Color code: White (general use), Yellow (caution), Green (safety inspector), Blue (electrical worker), Red (firefighter)
2. Safety Helmets: Protect the head from falling objects, electrical hazards, and other impacts.
 - Color code: Same as hard hats

Eye and Face Protection



1. Safety Glasses: Protect the eyes from chemical splashes, dust, and other hazards.
 - Color code: Clear (general use), Yellow (anti-fog), Gray (UV protection)
2. Goggles: Protect the eyes from chemical splashes, dust, and other hazards.
 - Color code: Same as safety glasses

Hearing Protection

1. Earplugs: Protect the ears from loud noises.
 - Color code: Yellow (foam earplugs), Orange (pre-molded earplugs)
2. Earmuffs: Protect the ears from loud noises.
 - Color code: Same as earplugs

Respiratory Protection



1. Dust Masks: Protect the lungs from dust, pollen, and other airborne particles.
 - Color code: White (general use), Yellow (dust mask with exhalation valve)
2. Respirators: Protect the lungs from hazardous chemicals, gases, and particles.

- Color code: Various colors depending on the type of respirator and its intended use

Hand Protection



1. Gloves: Protect the hands from cuts, abrasions, chemicals, and other hazards.
 - Color code: Various colors depending on the type of glove and its intended use (e.g., yellow for general use, green for medical use)

Foot Protection



1. Safety Shoes: Protect the feet from heavy objects, electrical hazards, and other impacts.
 - Color code: Various colors depending on the type of shoe and its intended use (e.g., black for general use, yellow for construction)

Body Protection

1. Coveralls: Protect the body from chemicals, dust, and other hazards.
 - Color code: Various colors depending on the type of coverall and its intended use (e.g., white for general use, yellow for hazardous materials)
2. Vests: Protect the torso from impacts, cuts, and other hazards.
 - Color code: Various colors depending on the type of vest and its intended use (e.g., orange for traffic control, yellow for construction)

safety signs and their meanings used in welding workshops:

common safety signs used in welding workshops, along with their meanings and explanations:

INFORMATIONAL SIGNS

1. Fire Extinguisher Location

- Meaning: Indicates the location of fire extinguishers.
 - Explanation: Fire extinguishers are essential for putting out fires quickly and effectively.
2. Emergency Phone Number
 - Meaning: Indicates the emergency phone number to call in case of an accident.
 - Explanation: Having the emergency phone number readily available can help ensure prompt medical attention in case of an accident.
 3. Welding Safety Procedures
 - Meaning: Indicates the welding safety procedures to follow.
 - Explanation: Welding safety procedures can help prevent accidents and ensure a safe working environment.

COLOR-CODED SIGNS:

1. Red - Danger, warning, or prohibition.
2. Yellow - Caution or warning.
3. Green - Safety or information.
4. Blue - Mandatory or instructional

PROHIBITION SIGNS

Prohibition signs are there to tell you not to do something.

They are a danger sign, telling you that it is not safe to proceed.

These signs have a red circle with a red diagonal line through it. The pictogram is black on a white background.

An easy way to remember the shape of prohibition signs is that it is a circle, like a full stop. And the sign literally means, stop.

1. No Open Flames



- Meaning: Indicates that open flames are not allowed in the area.
 - Explanation: Open flames can ignite flammable materials and cause fires.
2. No Smoking
 - Meaning: Indicates that smoking is not allowed in the area.
 - Explanation: Smoking can ignite flammable materials and cause fires.
 3. No Entry
 - Meaning: Indicates that entry is not allowed in the area.
 - Explanation: Certain areas in the welding workshop may be restricted due to safety hazards or ongoing welding processes.

Prohibition signs			
picture	colour	shape	example
	red	circle	
	Examples		
	meaning		No entry
	Do not stop	Do not operate	No parking No smoking No unauthorised persons

MANDATORY SIGNS

Mandatory signs are there to tell you to do something. Consider these the opposite of prohibition signs.

Mandatory signs are round. The pictogram is white on a blue background, and the circle usually has a white border.

Instead of being round for full stop (like prohibition signs), you can remember that mandatory signs are round for 'Obey'. Because that is what you must do, obey them.

1. Wear Protective Clothing

- Meaning: Indicates that protective clothing is required to prevent injury.
- Explanation: Protective clothing, such as gloves, safety glasses, and welding helmets, can prevent injury from sparks, heat, and other hazards.

2. Use Respiratory Protection

- Meaning: Indicates that respiratory protection is required to prevent inhalation of hazardous substances.
- Explanation: Welding processes can generate hazardous fumes and particles that can be inhaled.

Mandatory signs			
picture	colour	shape	example
	Blue	Circle	
	Examples		
	meaning	Keep clear	Wear a face mask
	Do this You must obey	Ear protection must be worn	Keep the fire door closed

WARNING SIGNS

Warning signs are there to warn you of dangers.

Unlike the two previous signs, they don't tell you not to do something, or to do something. They simply aim to make you aware of a danger, so you can protect yourself.

These signs are triangular. The pictogram is black on a yellow background, and the triangle has a black border.

1. Warning: Hot Surface



- Meaning: Indicates that a surface is hot and can cause burns.
- Explanation: Welding equipment, such as welding machines and heat exchangers, can become extremely hot during use.

2. Warning: Eye Protection Required



- Meaning: Indicates that eye protection is required to prevent eye damage.
- Explanation: Welding processes can emit intense light, sparks, and debris that can cause eye damage.

3. Warning: Hearing Protection Required



- Meaning: Indicates that hearing protection is required to prevent hearing damage.
- Explanation: Welding processes can generate loud noises that can cause hearing damage.

Warning signs			
picture	colour	shape	example
	Yellow	Triangle	
		Examples	
	meaning	Caution hot water	Danger 440 volts

	Warning, Be careful Be aware	Fragile roof	Mind the step Caution slippery floor
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Safe condition signs

Safe condition signs can usually be followed to safety.

And what colour better represents safety than green? That's right, they are green!

These signs are rectangular or square. The pictogram is white on a green background, and will often have a white border.

Safe condition signs			
picture	colour	shape	example
	Green	Rectangle	
		Examples	
	meaning	Fire exit	
	Follow me This way. Escape route First aid	Push bar to open	Assembly point First aid box Emergency stop

Fire Equipment Signs

Fire equipment/fighting signs show you where fire equipment is.

What colour do we think of for fire? Red. But we already used red, with the prohibition sign. Well, fire equipment signs are also red - but in a different shape.

Fire equipment/fighting signs are square or rectangular. The pictogram is white on a red background, and will often have a white border.

picture	colour	shape	example
	red	square	
		Examples	
	Meaning	Fire alarm call point	Fire alarm control panel Fire extinguisher Gas shut-off valve
	Here it is		For fire use only

FIRE EXTINGUISHING:

- Know how to use fire extinguishers
- Keep sand or fire extinguishing material nearby

Fire Hoses (Fire Safety Equipment)



1. Fire Extinguisher Types:



- Class A (ordinary combustibles)
- Class B (flammable liquids)
- Class C (electrical fires)

2. Fire Extinguisher Colors:

- Red (Class A)
- Yellow (Class B)
- Blue (Class C)

ACTIVITY 1

Section A (12 Marks)

Multiple-choice questions

Safety Procedures

1. What should you do in case of an emergency?
A) Call a friend
B) Call emergency services
C) Try to handle it alone
D) Panic
2. What is the purpose of a buddy system?
A) To have someone to talk to
B) To share workload
C) To ensure someone knows your location
D) To compete with others
3. What is a basic item in a first aid kit?
A) Band-Aids
B) Pain relievers
C) Antiseptic wipes
D) All of the above
4. What is the primary purpose of a first aid kit?
A) To treat serious injuries
B) To provide pain relief
C) To clean wounds
D) To provide basic care until medical help arrives
5. What color represents caution or warning?
A) Red
B) Yellow
C) Green
D) Blue
6. What type of clothing is essential for outdoor activities?
A) Hard hat
B) Safety glasses
C) Steel-toed boots
D) All of the above
7. What does a red and white sign indicate?
A) Danger

- B) Caution
- C) Information
- D) Safety

8. What type of sign provides information about trails?

- A) Warning sign
- B) Information sign
- C) Danger sign
- D) Directional sign

9. What is a common fire extinguishing material?

- A) Water
- B) Sand
- C) Fire extinguisher
- D) All of the above

10. What type of fire extinguisher is used for electrical fires?

- A) Class A
- B) Class B
- C) Class C
- D) Class D

11. What type of sign indicates danger?

- A) Yellow and black
- B) Red and white
- C) Green and white
- D) Blue and white

12. What material is commonly used to extinguish fires?

- A) Water
- B) Sand
- C) Fire extinguisher
- D) All of the above

Section B (20)

1. What are the essential items to include in a first aid kit?

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.....

2. What is the purpose of wearing protective clothing during outdoor activities?

.....

.....

3. What does a red and white safety sign typically indicate?

.....

4. What steps should be taken in case of a fire emergency

.....

.....

5. What is the importance of having a buddy system during outdoor activities?

.....

.....

6. What is the primary goal of safety procedures?

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.....

7. What type of clothing is essential for outdoor activities?

.....

.....

8. What does a yellow safety sign typically indicate?

.....

.....

9. What is the purpose of a fire extinguisher?

.....

.....

10. What should be done after treating a minor injury?

.....

.....

PERSONAL PROTECTIVE EQUIPMENT (PPE) THAT A LEARNER SHOULD USE WHEN DOING WELDING IN A WORKSHOP:

Head Protection:



1. Welding helmet (shielding from sparks, UV radiation, and heat)
2. Hard hat (protecting against falling objects)

Eye Protection:

Welding goggles (protecting eyes from sparks, debris, and UV radiation)

Welding helmets and goggles with UV-protective lenses are designed to protect the eyes from the intense ultraviolet (UV) radiation emitted during welding processes, such as arc welding, gas welding, and cutting. These lenses absorb or block UV radiation, preventing eye damage and discomfort.

Types of UV-protective lenses include:

- Shade 10-14 lenses for arc welding
- Shade 5-8 lenses for gas welding and cutting
- Specialized lenses for specific welding processes, such as laser welding or plasma cutting

Face Protection:

1. Face shield (protecting skin from sparks and heat)
2. Respiratory mask (protecting against fumes and particles)

Body Protection:

1. Welding jacket or apron (protecting clothing from sparks and heat)
2. Fire-resistant pants and shirt
3. Steel-toed boots (protecting feet from heavy objects)
4. Insulated gloves (protecting hands from heat, sparks, and electrical shock)

Respiratory Protection:

1. Ventilation system (removing fumes and particles)
2. Respiratory mask or respirator (filtering airborne contaminants)

Hearing Protection:

1. Earplugs or earmuffs (reducing noise exposure)

Welding Process-Specific PPE:

1. Shielded Metal Arc Welding (SMAW): Welding helmet, gloves, safety glasses
2. Gas Metal Arc Welding (GMAW): Welding helmet, gloves, safety glasses, respirator
3. Gas Tungsten Arc Welding (GTAW): Welding helmet, gloves, safety glasses, respirator

Remember to:

- Inspect PPE before use
- Properly maintain and store PPE
- Train learners on PPE usage and maintenance
- Ensure compliance with local regulations and industry standards

ACTIVITY 2

Personal Protective Equipment (PPE) for welding: (5 Marks)

1. What type of helmet is essential for welding?
A) Hard hat
B) Welding helmet
C) Safety glasses
D) Respirator

2. Which of the following protects eyes from UV radiation during welding?
A) Safety glasses
B) Goggles
C) Shade 10-14 lens
D) All of the above

3. What type of clothing is recommended for welding?
A) Cotton
B) Polyester
C) Fire-resistant
D) Short sleeves

4. Which PPE protects hands from heat, sparks, and electrical shock?
A) Insulated gloves
B) Safety glasses
C) Welding helmet
D) Steel-toed boots

5. What type of respiratory protection is necessary for welding?
A) Ventilation system
B) Respiratory mask
C) Face shield
D) All of the above



**SAFETY, GOOD HOUSEKEEPING, AND GENERAL
WORKSHOP RULES:**

Safety Rules

1. Personal Protective Equipment (PPE): Wear required PPE, such as safety glasses, gloves, and a face mask, when working with hazardous materials or equipment.
2. Fire Safety: Ensure that fire extinguishers are easily accessible and that you know how to use them. Never use water to extinguish electrical or oil fires.
3. Electrical Safety: Turn off equipment and machinery when not in use. Avoid overloading electrical outlets, and keep cords and cables organized.
4. Chemical Safety: Handle chemicals with care, and always follow the manufacturer's instructions. Wear protective gear, such as gloves and goggles, when handling chemicals.
5. Ladder Safety: Inspect ladders before use, and ensure they are securely positioned. Always maintain three points of contact (two hands and one foot, or two feet and one hand).

Good Housekeeping Rules

1. Cleanliness: Keep the workshop clean and tidy, including floors, workbenches, and equipment.
2. Organization: Label and organize tools, materials, and equipment to prevent loss and ensure easy access.
3. Waste Disposal: Dispose of waste properly, including recyclables, hazardous materials, and general trash.
4. Storage: Store materials and equipment in designated areas, and ensure they are securely stacked and labeled.

General Workshop Rules

By following these safety, good housekeeping, and general workshop rules, you can help create a safe and productive work environment.

1. Attendance and Sign-in: Sign in and out of the workshop, and ensure you are accounted for during emergencies.
2. Supervision: Ensure you are supervised by a qualified instructor or supervisor when working with new equipment or processes.

3. Equipment Use: Use equipment only for its intended purpose, and follow the manufacturer's instructions.
4. Tool Care: Properly maintain and store tools to prevent damage and ensure they remain in good working condition.
5. Emergency Procedures: Familiarize yourself with emergency procedures, including evacuation routes, fire extinguisher locations, and first aid kits.
6. Respect and Communication: Treat others with respect, and communicate clearly and effectively to prevent accidents and ensure a smooth workflow.
7. No Horseplay: No running, jumping, or playing in the workshop. This can cause accidents and damage to equipment.
8. No Food or Drink: No eating or drinking in the workshop, except in designated areas. This can prevent contamination and damage to equipment.
9. No Cell Phone Use: No using cell phones in the workshop, except in emergency situations. This can prevent distractions and accidents.
10. No Unauthorized Equipment Use: Only use equipment that you have been trained on and authorized to use. This can prevent accidents and damage to equipment.
11. No Removing Safety Guards: Never remove safety guards from equipment. This can prevent serious injury or death.
12. No Overreaching: Never overreach when working with equipment. This can cause loss of balance and accidents.
13. No Wearing Loose Clothing: Wear tight-fitting clothing and tie back long hair. This can prevent clothing from getting caught in equipment.
14. No Working Alone: Always work with a buddy, especially when working with hazardous equipment or materials. This can ensure help is available in case of an emergency.
15. No Ignoring Safety Alarms: Never ignore safety alarms or warning lights. This can indicate a serious safety hazard.
16. No Failing to Report Incidents: Always report incidents, accidents, or near-misses to your supervisor or instructor. This can help prevent future incidents and ensure a safe working environment.

UNSAFE CONDITIONS

Definition: Conditions that can cause harm, injury, or damage.

examples and explanations of unsafe conditions:

SLIP, TRIP, AND FALL HAZARDS

1. Wet or oily floors: Spills or leaks can create slippery surfaces, increasing the risk of falls.

2. Cluttered walkways: Obstacles, such as tools or materials, can cause tripping hazards.
3. Uneven or damaged flooring: Cracks, holes, or uneven surfaces can lead to trips and falls.

FIRE HAZARDS

1. Flammable materials: Storing flammable materials, such as chemicals or rags, in hot or spark-prone areas can increase the risk of fires.
2. Overloaded electrical outlets: Using too many extension cords or overloading outlets can cause electrical fires.
3. Poorly maintained equipment: Failing to maintain equipment, such as faulty wiring or clogged ventilation, can increase the risk of fires.

Electrical Hazards

1. Exposed wiring: Frayed or exposed wiring can cause electrical shocks or fires.
2. Overhead power lines: Working near overhead power lines without proper precautions can increase the risk of electrical shocks.
3. Using damaged or faulty equipment: Using equipment with damaged cords, plugs, or other components can increase the risk of electrical shocks.

Chemical Hazards

1. Improper storage: Storing chemicals in incompatible containers or locations can increase the risk of chemical reactions or spills.
2. Inadequate ventilation: Failing to provide adequate ventilation when working with chemicals can increase the risk of inhalation or explosion.
3. Insufficient personal protective equipment (PPE): Failing to wear proper PPE, such as gloves or goggles, when handling chemicals can increase the risk of exposure.

Ergonomic Hazards

1. Poorly designed workstations: Workstations that are not designed to accommodate employees' physical needs can increase the risk of musculoskeletal disorders.
2. Insufficient lighting: Inadequate lighting can increase the risk of eye strain, headaches, and accidents.
3. Inadequate breaks: Failing to provide regular breaks can increase the risk of fatigue, decreased productivity, and accidents.

CONSEQUENCES:

1. Accidents



2. Injuries
3. Damage to equipment
4. Financial losses
5. Legal liabilities

Identifying Unsafe Conditions:

Identifying unsafe conditions is a critical step in maintaining a safe work environment.

STEPS TO HELP YOU IDENTIFY UNSAFE CONDITIONS:

Observations

1. Walk-through inspections: Regularly walk through the workplace to identify potential hazards.
2. Look for signs of wear and tear: Check equipment, tools, and infrastructure for signs of damage or deterioration.
3. Check for proper lighting: Ensure that the workplace is well-lit, and that lighting is adequate for the tasks being performed.

Hazard Recognition

1. Slip, trip, and fall hazards: Look for spills, uneven surfaces, and obstacles that could cause slips, trips, or falls.
2. Fire hazards: Identify potential fire sources, such as open flames, sparks, or hot surfaces.
3. Electrical hazards: Check for exposed wiring, overloaded outlets, or damaged electrical equipment.
4. Chemical hazards: Identify potential chemical hazards, such as spills, leaks, or inadequate ventilation.

Employee Feedback

1. Encourage employee reporting: Encourage employees to report any hazards or concerns they may have.
2. Conduct regular safety meetings: Hold regular safety meetings to discuss potential hazards and receive feedback from employees.

Use of Checklists and Inspections

1. Use safety checklists: Use safety checklists to identify potential hazards and ensure that equipment and facilities are properly maintained.
2. Conduct regular inspections: Conduct regular inspections of equipment, facilities, and work processes to identify potential hazards.

1. Regular inspections
2. Reporting hazards
3. Risk assessments
4. Employee training
5. Safety audits

Correcting Unsafe Conditions:

1. Repair or replace broken equipment
2. Clean up spills immediately
3. Install safety guards
4. Improve lighting

Implement safety protocols

By recognizing and addressing unsafe conditions, we can prevent accidents and ensure a safer working environment.

unsafe acts in a welding workshop, along with their explanations:

Personal Protective Equipment (PPE) Violations

1. Not wearing safety glasses:
Failing to wear safety glasses can result in eye injuries from sparks, slag, or other debris.
2. Not wearing gloves:
Failing to wear gloves can result in hand injuries from cuts, burns, or electrical shock.
3. Not wearing a welding helmet:
Failing to wear a welding helmet can result in eye injuries from UV radiation, as well as head and neck injuries from sparks or falling objects.

Equipment Misuse

1. Using damaged or worn-out equipment:
Using damaged or worn-out equipment can result in equipment failure, electrical shock, or other accidents.
2. Not following equipment instructions:

Failing to follow equipment instructions can result in equipment misuse, accidents, or injuries.

3. Overloading equipment:

Overloading equipment can result in equipment failure, electrical shock, or other accidents.

Work Area Hazards

1. Cluttered work area:

A cluttered work area can result in tripping hazards, fires, or other accidents.

2. Poor ventilation:

Poor ventilation can result in inhalation of hazardous fumes, particles, or gases.

3. Inadequate lighting:

Inadequate lighting can result in eye strain, accidents, or injuries.

Electrical Hazards

1. Using damaged or frayed cords:

Using damaged or frayed cords can result in electrical shock or fires.

2. Overloading electrical outlets:

Overloading electrical outlets can result in electrical shock, fires, or equipment damage.

Fire Hazards

1. Not storing flammable materials properly:

Not storing flammable materials properly can result in fires or explosions.

2. Not having a fire extinguisher nearby:

Not having a fire extinguisher nearby can result in fires spreading or getting out of control.

3. Not following hot work procedures:

Not following hot work procedures can result in fires or explosions.

4. Horseplay or distractions:

Horseplay or distractions can result in accidents, injuries, or equipment damage.

5. Not following safety procedures:

Not following safety procedures can result in accidents, injuries, or equipment damage.

6 Not reporting hazards or incidents:

Not reporting hazards or incidents can result in accidents, injuries, or equipment damage.

CONSEQUENCES: ACCIDENTS, INJURIES, OR DAMAGE TO EQUIPMENT OR PROPERTY.

consequences of unsafe acts:

Physical Injuries



1. Cuts and lacerations:

Not wearing gloves or safety glasses can result in cuts and lacerations from sharp objects or flying particles.

2. Burns:

Not wearing protective clothing or following proper welding procedures can result in burns from sparks, flames, or hot surfaces.

3. Broken bones:

Not using proper lifting techniques or wearing personal fall protection equipment can result in broken bones from falls or strains.

Health Problems

1. Respiratory problems:

Not wearing respiratory protection or working in poorly ventilated areas can result in respiratory problems from inhaling hazardous fumes or particles.

2. Hearing loss:

Not wearing hearing protection or working in noisy areas can result in hearing loss from prolonged exposure to loud noises.

3. Eye damage:

Not wearing eye protection or working with hazardous chemicals can result in eye damage from chemical splashes or UV radiation.

Equipment Damage

1. Equipment failure:

Not following proper maintenance procedures or using damaged equipment can result in equipment failure, leading to costly repairs or replacement.

2. Fire damage:

Not following proper hot work procedures or using damaged equipment can result in fires, leading to equipment damage or destruction.

3. Property damage:

Not following proper safety procedures or using damaged equipment can result in property damage, leading to costly repairs or replacement.

Environmental Damage



1. Chemical spills:

Not following proper chemical handling procedures or using damaged equipment can result in chemical spills, leading to environmental damage or contamination.

2. Air pollution:

Not following proper ventilation procedures or using damaged equipment can result in air pollution, leading to environmental damage or health problems.

3. Water pollution:

Not following proper chemical handling procedures or using damaged equipment can result in water pollution, leading to environmental damage or contamination.

Economic Consequences

1. Lost productivity:

Unsafe acts can result in lost productivity due to injuries, equipment damage, or environmental damage.

2. Medical costs:

Unsafe acts can result in medical costs for treating injuries or illnesses.

3. Workers' compensation claims:

Unsafe acts can result in workers' compensation claims, leading to increased insurance premiums and costs.

Reputational Damage

1. Loss of public trust:

Unsafe acts can result in loss of public trust, damaging the reputation of the organization.

2. Negative media attention:

Unsafe acts can result in negative media attention, damaging the reputation of the organization.

3. Regulatory penalties:

Unsafe acts can result in regulatory penalties, damaging the reputation of the organization.

ACTIVITY 3 (15 Marks)

**TOPIC OF SAFETY
GOOD HOUSEKEEPING AND WORKSHOP RULES:
SECTION A**

1. What is the primary purpose of Personal Protective Equipment (PPE)?

- A) To protect equipment
- B) To protect the environment
- C) To protect oneself from harm
- D) To improve productivity

2. What should you do if you notice a spill in the workshop?

- A) Ignore it
- B) Clean it up immediately
- C) Report it to someone else
- D) Warn others

3. What is the importance of good housekeeping in a workshop?

- A) To reduce costs
- B) To increase productivity
- C) To prevent accidents
- D) All of the above

4. What should you do if you see someone not following safety rules?

- A) Ignore them
- B) Warn them
- C) Report them
- D) Intervene

5. Why are safety signs important in a workshop?

- A) To provide directions
- B) To warn of hazards
- C) To identify equipment
- D) To decorate the area

Q6: What is the primary reason for keeping a workshop safe and clean?

- A) To reduce costs
- B) To improve productivity
- C) To prevent accidents and injuries
- D) To comply with regulations

Q7: What is an example of an unsafe condition in a workshop?

- A) Well-lit workspace
- B) Organized storage
- C) Slippery floor
- D) Properly maintained equipment

Q8: What should be done immediately after a spill in the workshop?

- A) Report it to supervisor
- B) Clean it up immediately
- C) Warn others
- D) Ignore it

Q9: Why is good housekeeping important in a workshop?

- A) To reduce costs
- B) To improve productivity
- C) To prevent accidents and injuries
- D) All of the above

Q10: What is the consequence of neglecting safety and cleaning in a workshop?

- A) Increased productivity
- B) Improved reputation
- C) Accidents and injuries
- D) Reduced costs

Q11: What should be done daily to maintain a safe and clean workshop?

- A) Inspect equipment
- B) Clean up spills
- C) Review safety procedures
- D) All of the above

Q12: Why is it important to label and store chemicals properly?

- A) To reduce costs
- B) To improve productivity
- C) To prevent accidents and injuries
- D) To comply with regulations

Q13: What is the benefit of maintaining a safe and clean workshop?

- A) Reduced risk of accidents
- B) Improved employee health
- C) Increased productivity
- D) All of the above

Q14: What should you do if you notice a colleague not following safety procedures?

- A) Ignore them
- B) Warn them
- C) Report them to supervisor
- D) Intervene and correct them

Q15: Why is regular maintenance of equipment important in a workshop?

- A) To reduce costs
- B) To improve productivity
- C) To prevent accidents and injuries
- D) All of the above

PROPER PROTOCOL FOR DEALING WITH VICTIMS OF HIV AND AIDS:

Precautions

1. Use Personal Protective Equipment (PPE): Wear gloves, masks, and eye protection when handling bodily fluids or providing care.
2. Handle Bodily Fluids with Care: Use disposable gloves and containers when handling bodily fluids, such as blood, semen, or saliva.
3. Clean and Disinfect: Clean and disinfect surfaces and equipment that come into contact with bodily fluids.

First Aid and Emergency Response

1. Bleeding: Wear gloves and apply pressure to the wound using a clean cloth or gauze. Avoid direct contact with bodily fluids.
2. Cuts and Punctures: Wear gloves and clean the wound with soap and water. Apply antibiotic ointment and cover with a bandage.
3. Medical Emergency: Call for medical assistance immediately if the person is experiencing severe symptoms, such as difficulty breathing or chest pain.

Confidentiality and Stigma Reduction

1. Maintain Confidentiality: Protect the person's confidentiality and avoid disclosing their HIV/AIDS status without their consent.

2. Use Non-Judgmental Language: Avoid using language that perpetuates stigma or discrimination against people living with HIV/AIDS.
3. Provide Support and Resources: Offer support and resources, such as counseling or medical referrals, to help the person manage their condition.

EXAMPLES

1. Scenario: A person with HIV/AIDS experiences a cut while working in a workshop.

Protocol:

Wear gloves, clean the wound with soap and water, and apply antibiotic ointment. Cover the wound with a bandage and provide medical attention if necessary.

2. Scenario: A person with HIV/AIDS requires first aid for a bleeding injury.

Protocol:

Wear gloves, apply pressure to the wound using a clean cloth or gauze, and avoid direct contact with bodily fluids.

Proper Disposal of Cleaning Materials:

1. Dispose of gloves, wipes, and other contaminated materials in biohazard bags
2. Seal bags securely
3. Label bags as infectious waste
4. Follow local regulations for disposal

ACTIVITY 4 (10 Marks)

Proper protocol for dealing with victims of HIV and AIDS:

Q1: What is the primary precaution when handling bodily fluids of HIV/AIDS patients?

- A) Wear a face mask
- B) Wear gloves
- C) Wear protective eyewear
- D) All of the above

Q2: Why is it important to wear gloves when handling bodily fluids?

- A) To prevent skin irritation
- B) To prevent transmission of HIV/AIDS
- C) To maintain hygiene
- D) All of the above

Q3: What should be done with contaminated materials?

- A) Dispose of them in regular trash

- B) Seal them in biohazard bags
- C) Wash them with soap and water
- D) Autoclave them

Q4: What is the proper way to dispose of sharps?

- A) Throw them away in regular trash
- B) Place them in a sharps container
- C) Bury them
- D) Burn them

Q5: Why is confidentiality important when dealing with HIV/AIDS patients?

- A) To prevent stigma
- B) To prevent transmission
- C) To maintain patient trust
- D) All of the above

Q6: What should be done immediately after exposure to bodily fluids?

- A) Wash hands with soap and water
- B) Report incident to supervisor
- C) Seek medical attention
- D) All of the above

Q7: What is the purpose of universal precautions?

- A) To prevent transmission of HIV/AIDS only
- B) To prevent transmission of all infectious diseases
- C) To maintain hygiene
- D) To reduce costs

Q8: How often should healthcare workers update their knowledge on HIV/AIDS?

- A) Annually
- B) Bi-annually
- C) Quarterly
- D) Regularly

Q9: What is the importance of proper cleaning and disinfection?

- A) To prevent transmission of HIV/AIDS
- B) To maintain hygiene
- C) To reduce costs
- D) All of the above

Q10: Why is it important to avoid sharing personal items with HIV/AIDS patients?

- A) To prevent transmission
- B) To maintain hygiene

- C) To reduce costs
- D) To prevent stigma

PROJECT: FOLDING TABLE (USING METAL)

<https://youtu.be/M6hLwRlrN1s>



Learning Outcomes:

1. Measure and calculate materials needed for a metalwork project
2. Estimate costs based on material prices
3. Create a budget for a project
4. Apply welding techniques to join metal components
5. **Demonstrate understanding of metal properties and selection**

Materials Needed:

1. Metal sheets (mild steel or aluminum)
2. Metal tubing (for legs and frame)
3. Welding rods
4. Screws
5. Hinges
6. Nails
7. Metal primer and paint (optional)

Step 1: Design and Planning

- Determine the size of the folding table: length, width, height (in mm)
- Choose the type of metal and materials
- Create a sketch or drawing of the design

Step 2: Measure and Calculate Materials

1. Metal sheets:
 - Top: length x width x thickness (e.g., 1200mm x 600mm x 1.5mm)
 - Shelf: length x width x thickness (e.g., 1200mm x 600mm x 1.5mm)
2. Metal tubing:
 - Legs: 4 x (height x diameter) (e.g., 4 x (750mm x 25mm))
 - Frame: perimeter of top x diameter (e.g., 2(1200mm + 600mm) x 25mm)
3. Welding rods:
 - Estimate amount needed based on joint design

4. Screws:
 - Count the number of screws needed for assembly
5. Hinges:
 - 2-4 hinges depending on design
6. Nails:
 - Not required (using welding instead)
7. Metal primer and paint (optional):
 - Estimate amount needed based on surface area

Step 3: Calculate Costs (Using South African Prices)

1. Research prices for materials in South African Rands:
 - Metal sheets (e.g., R150/ sheet)
 - Metal tubing (e.g., R50/ metre)
 - Welding rods (e.g., R20/ kg)
 - Screws (e.g., R10/ pack)
 - Hinges (e.g., R20/ hinge)
 - Metal primer and paint (e.g., R50/ can)
2. Calculate total cost:
 - Metal sheets: (number of sheets) x (R150/ sheet)
 - Metal tubing: (total metres) x (R50/ metre)
 - Welding rods: (total kg) x (R20/ kg)
 - Screws: (number of packs) x (R10/ pack)
 - Hinges: (number of hinges) x (R20/ hinge)
 - Metal primer and paint: (number of cans) x (R50/ can)
 - Total Cost: sum of all material costs

EXAMPLE CALCULATION:

- Folding table size: 1200mm x 600mm x 750mm
- Metal sheets:
 - Top: 1 sheet x R150/ sheet = R150
 - Shelf: 1 sheet x R150/ sheet = R150
- Metal tubing:
 - Legs: 4 x 0.75 metres x R50/ metre = R150
 - Frame: 2(1200mm + 600mm) x 25mm x R50/ metre = R300
- Welding rods: 2 kg x R20/ kg = R40
- Screws: 1 pack x R10/ pack = R10
- Hinges: 2 x R20/ hinge = R40
- Metal primer and paint: 1 can x R50/ can = R50 (optional)
- Total Cost: R150 + R150 + R150 + R300 + R40 + R10 + R40 + R50 = R890

COMPREHENSIVE GUIDE TO BUILDING A FOLDING TABLE USING METAL, WITH MEASUREMENTS IN MILLIMETERS:

Materials Needed:

- 50 x 50 x 6 mm steel tubing (for legs and frame)
- 25 x 25 x 6 mm steel tubing (for table top supports)
- 6 mm thick steel plate (for table top)
- 50 mm steel hinges
- 20 mm steel screws
- 10 mm steel nuts and bolts
- Metal primer and paint
- Safety gear (welding helmet, gloves, etc.)

Tools Needed:

- MIG welder
- Arc welder
- Drill press
- Hand drill
- Jigsaw or bandsaw
- Sander
- Wrenches and sockets
- Tape measure and square

Instructions:

Step 1: Cut and Prepare the Frame

1. Cut four pieces of 50 x 50 x 6 mm steel tubing to 800 mm length for the table legs.
2. Cut two pieces of 50 x 50 x 6 mm steel tubing to 1200 mm length for the table frame.
3. Cut two pieces of 25 x 25 x 6 mm steel tubing to 600 mm length for the table top supports.
4. Sand all cut edges to smooth them out.

Step 2: Assemble the Frame

1. Use a MIG welder to weld the legs to the frame, creating a rectangular shape.
2. Use a MIG welder to weld the table top supports to the frame, spaced 300 mm apart.
3. Use a wrench to tighten the nuts and bolts.

Step 3: Cut and Prepare the Table Top

1. Cut a piece of 6 mm thick steel plate to 1200 mm x 600 mm size for the table top.
2. Sand the cut edges to smooth them out.
3. Apply metal primer and paint to the table top.

Step 4: Assemble the Table Top

1. Place the table top onto the table top supports.
2. Use 20 mm steel screws to attach the table top to the supports.
3. Tighten the screws securely.

Step 5: Add the Hinges and Folds

1. Attach 50 mm steel hinges to the table legs and frame, allowing the table to fold up.
2. Use 10 mm steel nuts and bolts to attach the hinges to the legs and frame.
3. Test the table's folding mechanism to ensure it works smoothly.

Step 6: Finish and Inspect

1. Apply a final coat of metal primer and paint to the entire table.
2. Inspect the table for any defects or weaknesses.
3. Test the table's stability and folding mechanism.

Safety Considerations:

- Always wear safety gear, including a welding helmet, gloves, and safety glasses, when working with metal and welding equipment.
- Ensure the work area is well-ventilated and free from flammable materials.
- Use proper lifting techniques to avoid injury when handling heavy metal components.

Budgeting:

- Determine the total budget for the project
- Compare estimated costs to budget
- Adjust design or materials to fit budget constraints

ORTHOGRAPHIC PROJECTION AND GRAPHIC COMMUNICATION FOR WELDING LEARNERS:

Orthographic Projection

Definition:

Meaning of Orthographic Projection

Orthographic projection is a method of drawing an object by means of plane views. While isometric drawing looks at all the three views (or faces) of an object together, orthographic looks at the same three views separately but linked together. In orthographic projections, we can look at each view and see its "correct or true shape".

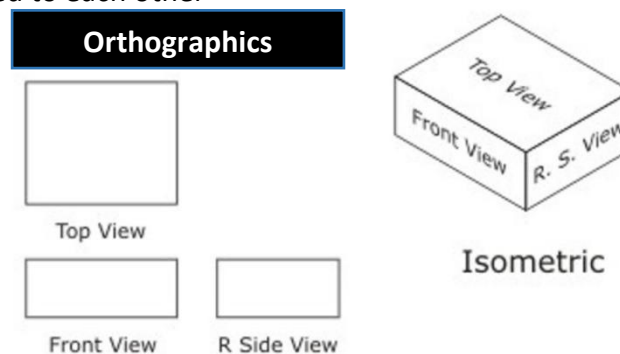
Orthographic projection is the only drawing that shows the shape, angles and size of each part of the article to be constructed as they really are. Because of this peculiar characteristic, it is one of the universally accepted drawings in the construction industry, therefore the drawing is usually used for production purposes.

Types of Projections:

1. Orthographic Projection (OP)
2. Isometric Projection (IP)
3. Oblique Projection (OB)

Orthographic Projection (OP)

1. Multi-view drawing: Front, Top, Side views
2. Each view shows 2D representation of 3D object
3. Views are aligned to each other



Isometric Projection (IP)

1. Single-view drawing showing 3D object
2. 120° angle between axes
3. Equal scaling in all directions

Oblique Projection (OB)

<https://youtu.be/ACHyDeFoILM>

1. Single-view drawing showing 3D object
2. 45° angle between axes
3. Unequal scaling in different directions

Drawing Symbols and Conventions:

1. Lines: Continuous, dashed, dotted
2. Dimensions: Size, location, orientation
3. Materials: Symbols for steel, aluminum, etc.
4. Welding symbols: Fillet, butt, groove welds

Graphic Communication in Welding

Importance:

1. Clear communication of design intent
2. Accurate interpretation of drawings
3. Efficient production planning

Drawing Requirements:

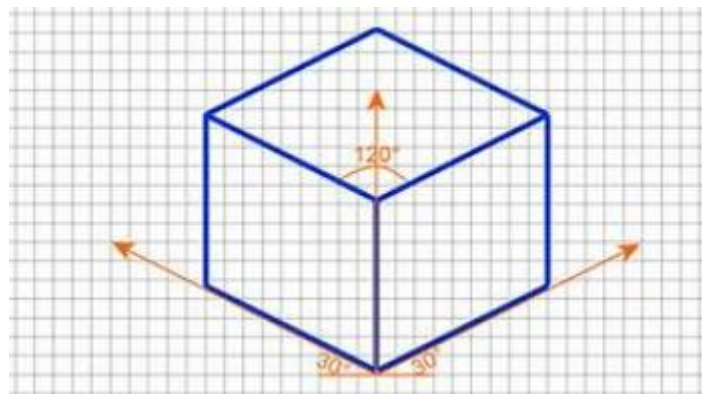
1. Scale: 1:1, 1:5, 1:10
2. Units: mm, cm, m
3. Tolerances: $\pm 1\text{mm}$, $\pm 0.5\text{mm}$
4. Material specifications
5. Welding symbols and instructions

Drawing Tools:

1. Graph paper
2. Pencils (HB, 2H)
3. Eraser
4. Ruler
5. Protractor
6. CAD software (optional)

ORTHOGRAPHIC PROJECTION OF A BLOCK:

Step 1: Determine the block's dimensions



- Length (L) = 50 mm
- Width (W) = 50 mm
- Height (H) = 20 mm

Step 2: Choose the projection views

- Front View (FV)
- Top View (TV)
- Right Side View (RSV)

Step 3: Draw the Front View (FV)

Draw a rectangle with dimensions: 50 mm x 20 mm
Label the dimensions

Step 4: Draw the Top View (TV)

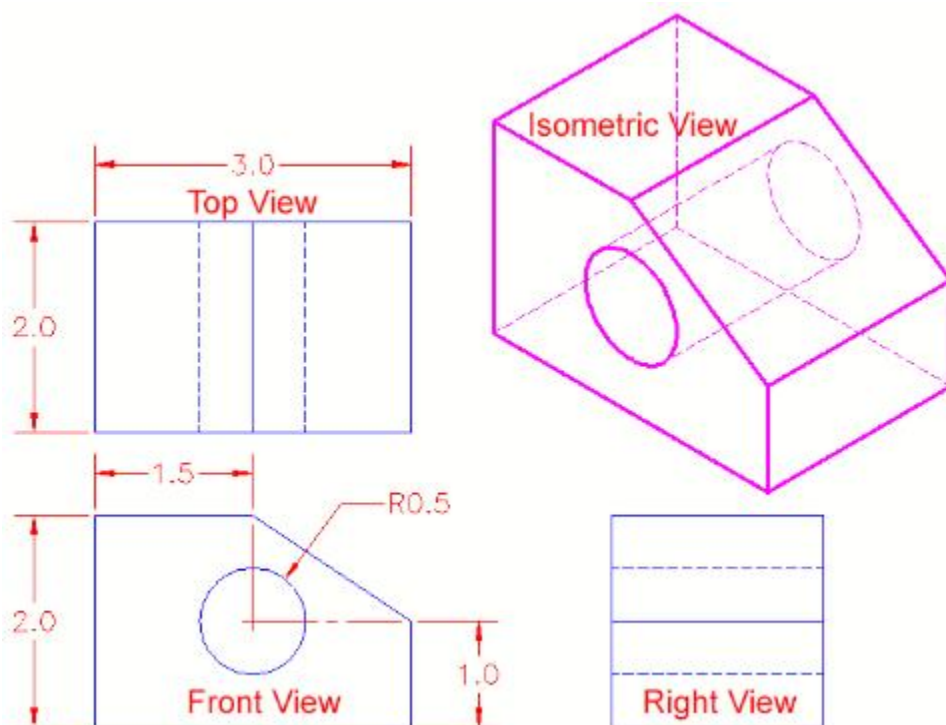
- Draw a square with dimensions: 50 mm x 50 mm
- Label the dimensions

Step 5: Draw the Right Side View (RSV)

- Draw a rectangle with dimensions: 50 mm x 20 mm
- Label the dimensions

Step 6: Align the views

- Ensure the views are aligned to each other
- Use vertical and horizontal lines to connect corresponding points

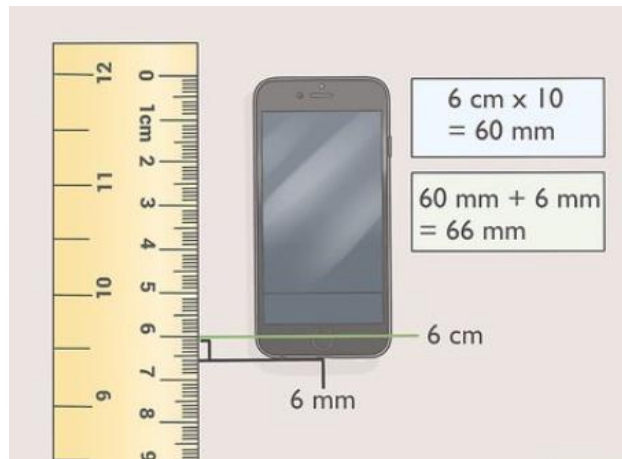


**MEASURING TOOLS AND THEIR PROPER USE FOR ACCURATELY MEASURING
LENGTH, WIDTH, AND THICKNESS IN MILLIMETERS:**

Measuring Tools

1. Ruler (Straight Edge)
2. Tape Measure (Flexible Ruler)
3. Vernier Caliper
4. Micrometer
5. Digital Caliper

Ruler (Straight Edge)



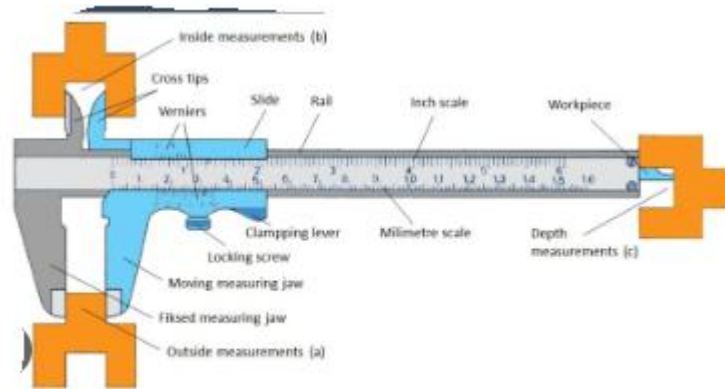
1. Used for measuring short lengths (up to 300mm)
2. Marked with millimeter and centimeter graduations
3. Ensure ruler is flat against surface

Tape Measure (Flexible Ruler)



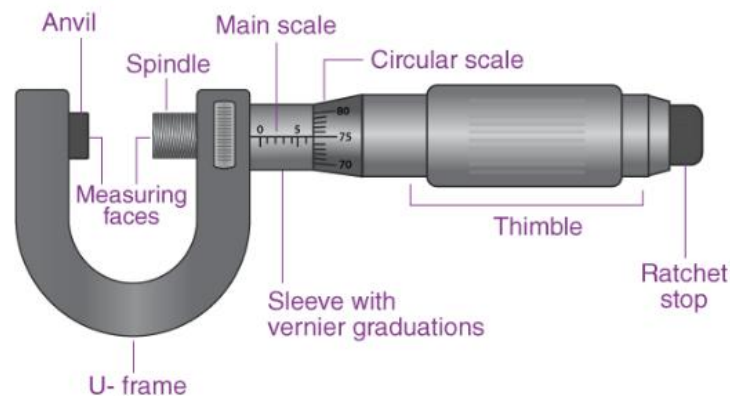
1. Used for measuring longer lengths (up to 10m)
2. Flexible, allowing for curved surfaces
3. Ensure tape is not twisted or bent

Vernier Caliper



1. Used for precise measurements ($\pm 0.1\text{mm}$)
2. Measures internal and external dimensions
3. Ensure jaws are closed gently

Micrometer



1. Used for highly precise measurements ($\pm 0.01\text{mm}$)
2. Measures small dimensions (e.g., wire thickness)
3. Ensure anvil and spindle are clean

Digital Caliper



1. Electronic measurement display
2. High accuracy ($\pm 0.01\text{mm}$)
3. Automatic unit conversion (mm)

Proper Measuring Techniques

1. Ensure measuring tool is clean and calibrated
2. Choose correct measuring tool for task

3. Measure multiple times to verify accuracy
4. Record measurements clearly

ACTIVITY 5

Questions related to measuring tools and techniques: (20 Marks)

Q1: What is the primary purpose of a vernier caliper?

.....

.....

Q2: What is the difference between a ruler and a tape measure?

.....

.....

Q3: How do you ensure accuracy when using a micrometer?

.....

.....

Q4: What is parallax error in measurement?

.....

.....

Q5: What is the benefit of using a digital caliper?

.....

.....

Q6: Why is it important to measure multiple times?

.....

.....

Q7: What is the correct technique for measuring thickness?

.....

.....

Q8: What safety precaution should be taken when handling measuring tools?

.....

.....

Q9: What is the purpose of zero-point setting on a measuring tool?

.....

.....

Q10: Why is proper measurement technique important?

.....

.....

***STEP-BY-STEP DEMONSTRATION ON HOW TO MEASURE AND READ IN
MILLIMETERS:***

MEASURING WITH A RULER

1. Place the ruler alongside the object to be measured.
2. Align the zero mark on the ruler with one end of the object.
3. Read the measurement at the other end of the object.

Reading Millimeter Measurements

1. Look for the millimeter markings on the ruler (usually small lines).
2. Count the number of millimeters from the zero mark.

Example Measurement

Object length: 56 mm

Ruler alignment:

| 0 mm | 10 mm | 20 mm | 30 mm | 40 mm | 50 mm | 56 mm |

Reading the Measurement

1. Align zero mark with object end.
2. Count 56 mm markings.

Measuring with a Vernier Caliper

1. Open the caliper jaws.

2. Place the object between the jaws.
3. Close the jaws until they touch the object.

Reading Vernier Caliper Measurements

1. Read the main scale (whole mm).
2. Read the vernier scale (0.1 mm increments).

Example Measurement

Object width: 32.5 mm

Main scale: 30 mm

Vernier scale: 2.5 mm

Reading the Measurement

1. Read main scale: 30 mm.
2. Read vernier scale: 2.5 mm.
3. Combine: 32.5 mm.

Digital Caliper Measurement

1. Place the object between the caliper jaws.
2. Close the jaws.

Reading Digital Caliper Measurements

1. Read the LCD display.

Example Measurement

Object thickness: 10.2 mm

LCD display: 10.2 mm

ACTIVITY 6

Questions and answers multiple choice (10 Marks)

Q1: What is the primary purpose of a vernier caliper?

- A) Measuring long lengths
- B) Measuring angles
- C) Measuring internal and external dimensions
- D) Marking straight lines

Q2: Which tool is used for transferring measurements or spacing equally?

- A) Tape measure
- B) Steel rule
- C) Dividers caliper
- D) Combination square

Q3: What feature does a high-tech combination square with digital display have?

- A) Automatic calculation of measurements
- B) Precision ruler
- C) Protractor for measuring angles
- D) All of the above

Q4: How often should measuring tools be calibrated?

- A) Daily
- B) Weekly
- C) Monthly
- D) Periodically

Q5: What is the benefit of using a marble top on a combination square?

- A) Increased precision
- B) Improved durability
- C) Enhanced visibility
- D) Reduced weight

Q6: Which tool is used for measuring long lengths?

- A) Steel rule
- B) Tape measure
- C) Vernier caliper
- D) Combination square

Q7: What is the primary purpose of a combination square?

- A) Measuring angles
- B) Marking straight lines
- C) Checking squareness
- D) All of the above

Q8: How should measuring tools be stored?

- A) In a dry, protected area
- B) In a humid environment
- C) In direct sunlight
- D) On the floor

Q9: What safety precaution should be taken when handling measuring tools?

- A) Wear protective gear
- B) Handle tools carelessly
- C) Use damaged tools
- D) Work in poor lighting

Q10: What is the benefit of using a digital caliper?

- A) Increased precision
- B) Improved durability
- C) Enhanced visibility
- D) Automatic calculation of measurements

PRACTICAL DEMONSTRATION GUIDE FOR LEARNERS TO DIFFERENTIATE BETWEEN NON-FERROUS AND FERROUS METALS:

✧ Materials Needed

- Ferrous metal samples (steel, iron)
- Non-ferrous metal samples (aluminum, copper, brass)
- Magnet
- Acid (vinegar or hydrochloric acid)
- Grinding wheel or spark tester
- File
- Density kit or calculator
- Protective gear (gloves, safety glasses)

By conducting these tests, learners can practically demonstrate their understanding of the differences between ferrous and non-ferrous metals.

❖ TEST 1: MAGNETIC TEST

1. Obtain samples of ferrous (e.g., steel) and non-ferrous (e.g., aluminum, copper) metals.
2. Hold a magnet close to each sample.
3. Observe the reaction:
 - Ferrous metals will attract the magnet.
 - Non-ferrous metals will not attract the magnet.

❖ TEST 2: ACID TEST (VINEGAR OR HYDROCHLORIC ACID)

1. Obtain samples of ferrous and non-ferrous metals.
2. Apply a few drops of acid (vinegar or hydrochloric acid) to each sample.
3. Observe the reaction:
 - Ferrous metals will show a bubbling or foaming reaction.
 - Non-ferrous metals will not react.

❖ TEST 3: SPARK TEST

1. Use a grinding wheel or spark tester to generate sparks on each metal sample.
2. Observe the sparks:
 - Ferrous metals will produce orange or yellow sparks.
 - Non-ferrous metals will produce white or silver sparks.

❖ TEST 4: FILING TEST

1. Use a file to scratch the surface of each metal sample.
2. Observe the filing:
 - Ferrous metals will produce a rough, brittle filing.
 - Non-ferrous metals will produce a smooth, soft filing.

METAL PROFILES USED IN WELDING:

Metal Profiles

1. Flat Bar
 - Used for: Brackets, frames, and supports
 - Characteristics: Flat, rectangular cross-section



2. Angle Iron
 - Used for: Framing, corners, and supports

- Characteristics: L-shaped cross-section



3. Channel Iron

- Used for: Framing, supports, and structural applications
- Characteristics: C-shaped cross-section



4. I-Beam

- Used for: Heavy structural applications, beams, and columns
- Characteristics: I-shaped cross-section



5. Pipe

- Used for: Plumbing, piping, and tubular structures
- Characteristics: Circular cross-section



5. Tube

- Used for: Structural applications, frames, and supports
- Characteristics: Hollow, circular or rectangular cross-section



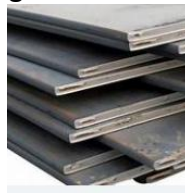
6. Sheet Metal

- Used for: Enclosures, panels, and covers
- Characteristics: Thin, flat, rectangular cross-section



8. Plate

- Used for: Heavy structural applications, bases, and supports
- Characteristics: Thick, flat, rectangular cross-section



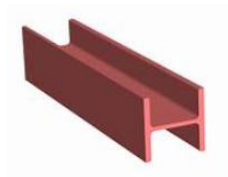
9. Rod

- Used for: Supports, brackets, and welding projects
- Characteristics: Round or square cross-section



10. H-Beam

- Used for: Heavy structural applications, beams, and columns
- Characteristics: H-shaped cross-section



Material Types

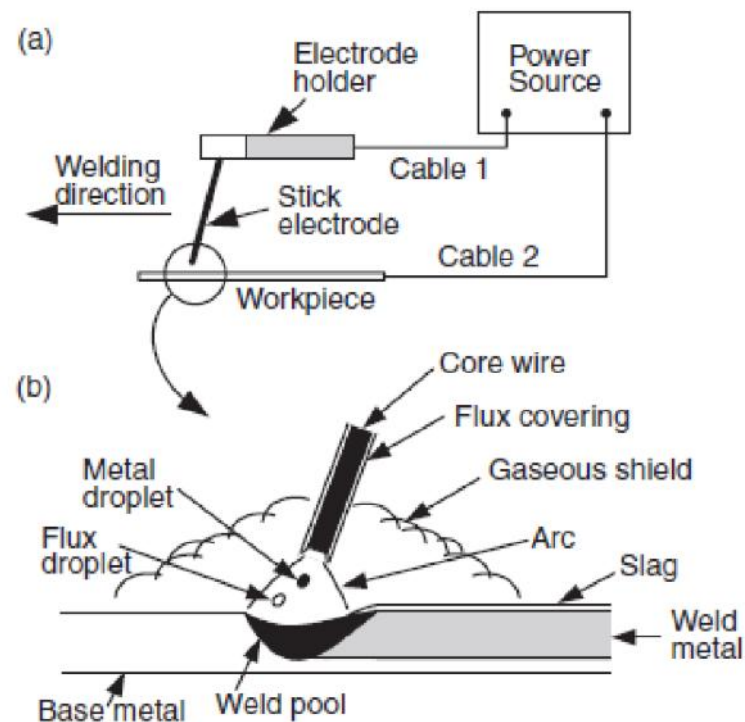
	Characteristics /Properties	uses	industries
Mild Steel	Low carbon content, easy to weld		
Stainless Steel	Corrosion-resistant, high temperature resistance		
Aluminum	Lightweight, corrosion- resistant	Aerospace, transportation, packaging	Aviation, automotive, food

			packaging
Copper	High electrical conductivity, corrosion-resistant	Electrical wiring, plumbing, electronics	Electrical, construction, electronics
Brass	Corrosion-resistant, durable	Plumbing, hardware, musical instruments	Plumbing, construction, music

Importance of Material Selection

1. Ensures structural integrity and safety.
2. Optimizes performance and efficiency.
3. Reduces maintenance and replacement costs.
4. Enhances product lifespan and durability.
5. Meets industry standards and regulations.

SHIELDED METAL ARC WELDING (SMAW) PROCESS, ALSO KNOWN AS "STICK" WELDING:



Shielded Metal Arc Welding (SMAW)

Definition: Shielded Metal Arc Welding (SMAW) is a manual welding process that uses a consumable electrode covered in a flux to protect the arc and molten metal from atmospheric contamination.

Equipment:

1. Welding machine (DC or AC)
2. Electrode holder
3. Shielded metal arc welding electrodes (SMAW electrodes)
4. Welding helmet
5. Safety glasses
6. Welding gloves
7. Fire-resistant clothing

Process:

1. Prepare the workpiece: Clean and deburr the metal surfaces.
2. Select the electrode: Choose the correct electrode diameter and type for the metal being welded.
3. Set up the welding machine: Adjust the machine settings according to the electrode and metal type.
4. Strike the arc: Hold the electrode at a 45-degree angle and strike the arc by scratching the electrode against the workpiece.
5. Maintain the arc: Hold the electrode at the correct angle (10-20 degrees) and maintain the arc length (1/8 inch).
6. Weld: Move the electrode in a steady, consistent motion along the joint.
7. Complete the weld: Release the arc and inspect the weld.

Shielded Metal Arc Welding Electrodes:

1. Covered electrodes: Coated with flux to protect the arc and molten metal.
2. Electrode classification: Identified by a four- or five-digit code (e.g., E6010).

Advantages:

1. Low equipment cost
2. Portable and versatile
3. Easy to learn
4. Effective for thick metals
5. Good for outdoor welding

Disadvantages:

1. Limited precision
2. Slow welding speed
3. Requires frequent electrode changes
4. Produces slag and spatter

5. Not suitable for thin metals or delicate work

Applications:

1. Construction
2. Heavy equipment repair
3. Shipbuilding
4. Pipe welding
5. Maintenance and repair

ACTIVITY 7

Section A (9 Marks)

MULTIPLE-CHOICE QUESTIONS AND ANSWERS ON SHIELDED METAL ARC WELDING (SMAW):

Q1: What is the primary purpose of the flux in SMAW electrodes?

- A) To increase welding speed
- B) To protect the arc and molten metal from atmospheric contamination
- C) To reduce electrode cost
- D) To improve weld appearance

Q2: Which of the following metals cannot be welded using SMAW?

- A) Carbon steel
- B) Alloy steel
- C) Stainless steel
- D) Aluminum

Q3: What is the typical angle for holding the SMAW electrode?

- A) 45-60 degrees
- B) 10-20 degrees
- C) 30-40 degrees
- D) 60-90 degrees

Q4: What is the primary disadvantage of SMAW?

- A) High equipment cost
- B) Limited precision
- C) Produces slag and spatter
- D) Requires specialized training

Q5: What is the classification system for SMAW electrodes?

- A) AWS (American Welding Society)
- B) ASME (American Society of Mechanical Engineers)
- C) Four- or five-digit code
- D) Alphabetical code

Q6: What safety precaution should be taken when performing SMAW?

- A) Wear protective gear (helmet, gloves, safety glasses)
- B) Ensure proper ventilation
- C) Avoid electrical shock
- D) All of the above

Q7: What is the purpose of the slag in SMAW?

- A) To protect the weld from atmospheric contamination
- B) To improve weld appearance
- C) To increase welding speed
- D) To reduce electrode cost

Q8: Which of the following is an advantage of SMAW?

- A) High equipment cost
- B) Limited precision
- C) Portable and versatile
- D) Requires specialized training

Q9: What is the recommended electrode size for SMAW?

- A) 1/8 inch (3.2 mm)
- B) 1/4 inch (6.4 mm)
- C) 1/2 inch (12.7 mm)
- D) Depends on the application

SECTION B (22 Marks)

QUESTIONS AND ANSWERS ON SHIELDED METAL ARC WELDING (SMAW):

Q1: What is Shielded Metal Arc Welding (SMAW)?

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Q2: What is the purpose of the flux in SMAW electrodes?

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Q3: What types of metals can be welded using SMAW?

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Q4: What is the advantage of using SMAW for outdoor welding?

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Q5: What is the primary disadvantage of SMAW?

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Q6: How often should SMAW electrodes be changed?

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Q7: What is the correct angle for holding the SMAW electrode?

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Q8: What safety precaution should be taken when performing SMAW?

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Q9: What is the classification system for SMAW electrodes?

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Q10: Can SMAW be used for thin metals?

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Q11: What is the purpose of the slag in SMAW?

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Pre-Operational Checks:

Step 1: Visual Inspection (10 minutes)

1. Check welding machine for damage or wear.
2. Verify cable and connection integrity.
3. Inspect electrode holder and cable.
4. Check workpiece lead and connections.

Step 2: Electrical Checks (5 minutes)

1. Ensure proper grounding.
2. Verify welding machine voltage and amperage settings.
3. Check electrical connections (tightness and security).

Step 3: Equipment Setup (10 minutes)

1. Connect electrode holder to welding machine.
2. Attach workpiece lead to workpiece.
3. Set welding machine settings (voltage, amperage, polarity).
4. Connect shielding gas (if applicable).

Step 4: Safety Checks (5 minutes)

1. Ensure proper ventilation.
2. Check fire extinguisher availability.
3. Verify personal protective equipment (PPE).

Welding galvanized materials poses unique safety risks due to the release of toxic fumes and particles. Proper ventilation is crucial to minimize exposure:

Risks associated with welding galvanized materials:

1. Zinc oxide fumes: Galvanized coatings release zinc oxide when heated, causing:
 - Respiratory issues (inflammation, bronchitis)
 - Metal fume fever (headaches, fever, chills)
2. Particulate matter: Welding galvanized materials generates airborne particles, including:
 - Zinc particles
 - Iron oxide
 - Other metallic particles
3. Volatile organic compounds (VOCs): Galvanized materials may contain VOCs, which can cause:
 - Respiratory problems
 - Eye irritation
 - Skin problems

Importance of ventilation:

1. Removes toxic fumes and particles: Ventilation helps extract hazardous substances from the air.
2. Reduces exposure risks: Proper ventilation minimizes welder exposure to toxic substances.
3. Prevents accumulation: Ventilation prevents the buildup of hazardous substances in the work area.

Ventilation requirements:

1. Local exhaust ventilation (LEV): Use LEV systems to capture fumes and particles at the source.
2. General ventilation: Ensure good airflow in the work area (at least 10 air changes per hour).
3. Respiratory protection: Wear respirators or masks with HEPA filters when welding galvanized materials.

Best practices:

1. Use a fume extractor or ventilation system specifically designed for welding.
2. Position ventilation equipment to capture fumes and particles effectively.
3. Regularly inspect and maintain ventilation equipment.
4. Ensure adequate airflow in the work area.
5. Monitor air quality and adjust ventilation accordingly.

Consequences of inadequate ventilation:

1. Health problems (respiratory issues, metal fume fever)
2. Increased risk of accidents (reduced visibility, impaired judgment)
3. Decreased productivity (worker fatigue, absenteeism)
4. Non-compliance with regulatory guidelines (fines, penalties)

step-by-step guide on replacing broken welding helmet glasses and adjusting the headband:

Replacing Broken Welding Helmet Glasses:

Green Glasses (Shade 10-13):

1. Switch off the welding machine and ensure the area is safe.
2. Remove the helmet from your head.
3. Locate the glass retention clips or screws.
4. Release the clips or remove the screws.
5. Carefully lift out the broken green glass.

6. Inspect the helmet's lens holder for debris.
7. Place the new green glass into the lens holder.
8. Secure the glass with clips or screws.

Clear Glasses (Shade 2-3):

1. Follow steps 1-4 above.
2. Remove the clear glass lens from the helmet.
3. Inspect the lens holder for debris.
4. Place the new clear glass lens into the holder.
5. Secure the lens with clips or screws.

Adjusting the Headband:

1. Put on the welding helmet.
2. Adjust the headband length by sliding the adjustment mechanism.
3. Tighten the headband by turning the adjustment knob.
4. Ensure a comfortable, secure fit.
5. Adjust the forehead pad for optimal comfort.

Proper Adjustment:

1. Helmet should sit level on the head.
2. Headband should be snug but not too tight.
3. Forehead pad should be centered.
4. Chin strap (if present) should be adjusted for comfort.

Demonstration:

Visual demonstration of replacing broken glasses and adjusting the headband

the correct procedures for welding galvanized materials:

Introduction to Entrepreneurship

- Definition: Entrepreneurship is the process of designing, launching, and running a business.
- Importance: Entrepreneurship creates jobs, drives innovation, and contributes to economic growth.

Essential Skills for Entrepreneurs

1. Willingness to Learn: Adaptability, curiosity, and openness to new ideas.
2. Ability to Listen: Effective communication, feedback, and customer understanding.
3. Creativity: Innovative thinking, problem-solving, and idea generation.

4. Inventiveness: Developing unique solutions and products.
5. Confidence: Self-assurance, resilience, and decision-making.
6. Perseverance: Overcoming obstacles, persistence, and determination.
7. Courage: Risk-taking, bold decision-making, and uncertainty management.
8. Risk Management: Assessing and mitigating risks.
9. Financial Management: Budgeting, forecasting, and cash flow management.
10. Leadership: Team building, motivation, and strategic planning.

Key Characteristics of Successful Entrepreneurs

1. Passion and motivation
2. Visionary thinking
3. Flexibility and adaptability
4. Resilience and stress management
5. Strong work ethic
6. Strategic planning
7. Networking and collaboration
8. Continuous learning
9. Innovative problem-solving
10. Customer-centric mindset

Steps to Become an Entrepreneur

1. Identify Your Passion: Find a business idea you're passionate about.
2. Conduct Market Research: Understand your target market and competition.
3. Develop a Business Plan: Outline goals, strategies, and financial projections.
4. Secure Funding: Explore financing options (loans, grants, investors).
5. Build a Team: Hire skilled and dedicated employees.
6. Launch and Iterate: Launch your business and continuously improve.
7. Monitor and Evaluate: Track progress, adjust strategies, and measure success.

Challenges Faced by Entrepreneurs

1. Financial constraints
2. Market competition
3. Regulatory compliance
4. Talent acquisition and retention
5. Risk management
6. Scaling and growth
7. Time management
8. Self-doubt and burnout

Best Practices for Entrepreneurs

1. Stay focused on your vision
2. Continuously innovate and improve
3. Build strong relationships with customers and partners

4. Foster a positive company culture
5. Stay adaptable and resilient
6. Prioritize financial management
7. Seek mentorship and guidance
8. Celebrate successes and learn from failures

Resources for Entrepreneurs

1. Small Business Administration (SBA)
2. Entrepreneurship courses and workshops
3. Business incubators and accelerators
4. Mentorship programs
5. Online business communities and forums
6. Business plan templates and tools
7. Industry-specific associations and conferences

Assessment Criteria for Entrepreneurship

1. Business plan development
2. Market research and analysis
3. Financial planning and management
4. Team building and leadership
5. Innovation and creativity
6. Risk management and adaptability
7. Communication and presentation skills
8. Problem-solving and decision-making

These notes provide a comprehensive overview of the essential skills, characteristics, and steps required to become a successful entrepreneur.

As a South African citizen, here are the key documentations required to become an entrepreneur:

Business Registration Documents:

1. Business Name Registration: Register your business name with the Companies and Intellectual Property Commission (CIPC).
2. Company Registration: Register your company with CIPC (e.g., Pty Ltd, CC, Sole Proprietor).
3. Business License: Obtain a business license from your local municipality.
4. Tax Registration: Register for taxes with the South African Revenue Service (SARS).
5. VAT Registration: Register for Value-Added Tax (VAT) if your annual turnover exceeds R1 million.

Personal Documents:

1. ID Document: Valid South African ID book or Smart ID card.

2. Passport: Valid South African passport (for international travel).
3. Proof of Residence: Utility bill or lease agreement.

Business Plan and Financial Documents:

1. Business Plan: Comprehensive business plan outlining goals, strategies, and financial projections.
2. Financial Statements: Audited financial statements (e.g., balance sheet, income statement).
3. Bank Account: Open a business bank account.

Regulatory Compliance Documents:

1. B-BBEE Certificate: Broad-Based Black Economic Empowerment (B-BBEE) certification.
2. COIDA Registration: Compensation for Occupational Injuries and Diseases Act (COIDA) registration.
3. UIF Registration: Unemployment Insurance Fund (UIF) registration.
4. Skills Development Levy (SDL) Registration: SDL registration.

Industry-Specific Licenses and Permits:

1. Health and Safety Certificates: Obtain health and safety certificates (e.g., OHS Act).
2. Environmental Permits: Obtain environmental permits (e.g., waste management).
3. Industry-specific licenses: Obtain licenses specific to your industry (e.g., food, construction).

Funding and Support Documents:

1. Funding Applications: Apply for funding from institutions (e.g., NSFAS, SEDA).
2. Business Incubator Applications: Apply for business incubator programs.
3. Mentorship Programs: Apply for mentorship programs.

Useful Resources:

1. Companies and Intellectual Property Commission (CIPC): (link unavailable)
2. South African Revenue Service (SARS): (link unavailable)
3. Small Enterprise Development Agency (SEDA): (link unavailable)
4. National Small Business Chamber (NSBC): (link unavailable)

Please note that documentation requirements may vary depending on your industry, location, and business type. Consult with relevant authorities and professionals to ensure compliance.